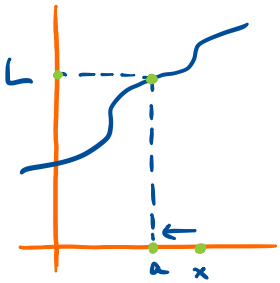
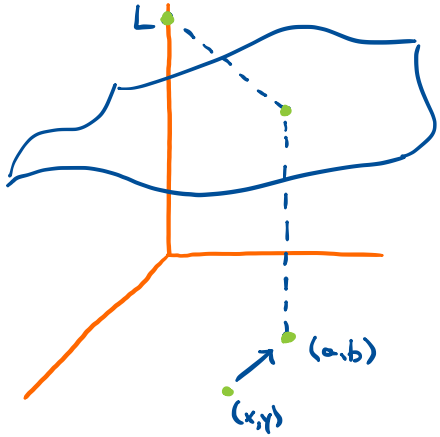
Calc I:



$$\lim_{x \to a} f(x) = L$$

Calc III:



$$\lim_{(x,y) \to (a,b)} f(x,y) = L$$

Fact. If there are two paths to (a,b) along which

$$\lim_{(x,y) \to (a,b)} f(x,y)$$

does not have the same value, then the limit does not exist.

→ Do problems 1,2.

$f(x,y)$ is continuous at (a,b) if

$$\lim_{(x,y) \to (a,b)} f(x,y) = f(a,b).$$

Polynomials, trig functions, exponentials, logarithms, and roots (where they are defined) are all continuous. So are sums, differences, products, and quotients (where they are defined) of them. → Do problems 3,4.